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FoodEx2 maintenance 2022

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Abstract

FoodEx2 is a comprehensive food classification and description system developed and maintained by EFSA to provide a standardized terminology backbone for representing and interpreting samples across the entire food safety risk assessment lifecycle, from data collection across different food and feed safety domains (e.g., monitoring of pesticides residues or biological hazards) to exposure assessment. To keep this terminology and sample coding framework fit-for-purpose, meeting evolving scientific and legislative requirements, a regular maintenance of this system is of utmost importance. Five major updates were carried out so far, with all performed changes documented in annual maintenance reports. This technical report describes the outcome of the sixth maintenance process done in 2022, which contained addition of new terms, inclusion of a novel hierarchy, reportability changes of some terms and amendments to existing terms including enrichment of implicit facets and changes of the hierarchical relationships.

Key words: food classification, food description, FoodEx2

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Summary

FoodEx2 is a standardised food classification and description system developed by EFSA to improve the comparability and exchange of data coming from different food safety domains within the remit of EFSA. In order to meet this requirement, it needs to be actively maintained, remain up to date as well as flexible to accommodate data collections in different domains. An annual maintenance of the system has taken place since April 2015 when the major revision (FoodEx2 revision 2) was performed.

A maintenance technical group composed of EFSA staff with expertise in different domains (food consumption, chemical contaminants, pesticides residues, etc.) was set-up to evaluate proposals and requests provided by users and stakeholders and take decisions regarding their implementation. The first maintenance was carried out in 2015 followed by five major releases performed between 2016 and 2022.

During the sixth maintenance process performed in 2022, the following major changes have been implemented: A new hierarchy, the PRIMo (Pesticide Residue Intake Model) hierarchy, has been incorporated with 1901 terms assigned. 137 new terms have been added to accommodate the needs of a new data collection (notably, a data collection on the migration of plasticisers in food contact materials) as well as existing data collections (such as the Avian Influenza monitoring programmes and the chemical monitoring data collection), or requests from other stakeholders – in this regard, 41 terms have been added based on requests from Food and Agricultural Organisation (FAO) / World Health Organization (WHO) for mapping the food consumption data within the Global Individual Food consumption data Tool – GIFT.

A systematic revision of the 'Bird (as animal)' section within the F01 – Source facet was concluded in 2022. The facet was restructured and updated to align with acknowledged, state-of-the-art scientific taxonomic classifications. 660 term descriptions have been improved or updated by revising their scope notes, which particularly affected the Botanicals hierarchy, i.e., the branch of the 'Plants (as plant)' section, where external references have been updated to link to the novel authoritative plant taxonomic database World Flora Online (WFO)[1]. In addition, the Feed hierarchy was revised according to the latest version of the Catalogue of Feed materials[2], with changes on 67 terms implemented in total. Among these changes, 33 new terms were added to the catalogue, while 17 terms were dismissed. Consequently, 17 terms were updated in terms of extended name, scope notes, parent-child relationship, implicit facet, or other implicit attributes.

The present report describes the outcome of the sixth maintenance process accomplished in 2022, to keep track of the changes in the system within two minor releases published in 2022 (MTX 13.1 and MTX 13.2) and the major release published in January 2023 (MTX 14.0).

[1] WFO (2023): World Flora Online. Published on the Internet; http://www.worldfloraonline.org. Accessed on: 27 Feb 2023
[2] Official Journal of the European Union, Commission Regulation (EU) 2022/1104 amending Regulation (EU) No 68/2013 on the Catalogue of feed materials, https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:32022R1104&from=EN

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1 Introduction

1.1 Background and terms of reference as provided by the requestor

In 2011, EFSA released a comprehensive food classification and description system named FoodEx2 (revision 1) (EFSA, 2011a, b). This system was developed with the aim of harmonising the description of the food matrices within data collections in different food safety domains. Following the publication, a testing phase was carried out in order to highlight strengths and weaknesses of the classification and to identify possible issues and needs for refinement. Based on the outcome of the testing phase, EFSA revised the system and in 2015 the second revision of the food classification and description system (FoodEx2 revision 2) was published (EFSA, 2015).

According to the guidance documents on FoodEx2 (EFSA, 2011a, 2015), internal and external users should provide to EFSA proposals for implementation and maintenance in order to refine the system and an annual EFSA Maintenance Technical Group (TG) should be set up in order to evaluate the suggestions and decide on their implementation. In fact, innovation in the food market and gained experience by using the system led to the need of keeping the catalogue updated. Following the revision 2 of the system, the FoodEx2 catalogue was revised five times based on EFSA needs and evaluation of the data providers and stakeholders' suggestions. A first maintenance took place in 2015 (EFSA et al., 2016) followed by four major releases performed between 2016-2021. The updated catalogues (MTX 10.0, MTX 11.0, MTX 12.0 and MTX 13.0) and the technical reports that summarize all changes performed during the maintenance from 2016 - 2021 were published (EFSA, 2019b, EFSA, 2020, EFSA, 2021b, EFSA, 2022).

Moreover, harmonized data collection in the areas of pesticides, contaminant occurrence, veterinary medicinal product residues and food additives (EFSA, 2021a) is reflected in the maintenance process of the FoodEx2 catalogue. Considering that all the above data domains and a number of other ad-hoc data collections in EFSA use the FoodEx2 catalogue, there was a need to accommodate different requests. Some changes were performed to enable giving a clear guidance (in terms of FoodEx2 codes) to data providers before the data collection launch date, while others were requested from EFSA staff working in different domains, data providers and network members (hereafter stated as users).

By using the system, both internal and external users and stakeholders provided constructive comments and requests for addition of missing terms and other type of updates in different sections of the catalogue. All these suggestions were considered for the maintenance purposes. A technical group composed of EFSA staff with expertise in different domains carried out the maintenance process evaluating whether and how to implement the received suggestions in the FoodEx2 system.

The present report describes the outcome of the sixth maintenance process undertaken in 2022 to keep track of the changes in the system implemented within two minor releases published in 2022 (MTX 13.1 and MTX 13.2) and the major release published in January 2023 (MTX 14.0). In order to perform a correct coding, with a reasonable level of detail, this report should be read in conjunction with the technical report 'The food classification and description system FoodEx2 (revision 2)' (EFSA, 2015) where the logic and the structure of the system and the rules for the correct use of FoodEx2 are described.

2 Data and Methodologies

2.1 Data

The FoodEx2 catalogue MTX 13.0 - version published in January 2022 was used as a starting point. The suggestions provided during the period between January 2022 and January 2023 by users and stakeholders of the FoodEx2 system were collected through written feedback and used to update the system.

2.2 Methodologies

Each request for addition or changes in the structure was considered and based on expert judgment, the TG decided whether a change was needed, or the request was already covered by the available terms and facet descriptors. All changes were made following the structure and the logic of the system as described in the report on the development of FoodEx2 revision 2 (EFSA, 2015).

2.2.1 Tools

The tools used for implementing the maintenance were Microsoft Excel® and the FoodEx2 Catalogue Browser (EFSA, 2019a), a user-friendly tool developed in Java by EFSA staff (Integrated Data former Evidence Management Unit) to help create FoodEx2 codes and to facilitate the management of the MTX catalogue. For the updating of external web links associated with the terms' scope notes, R (R Core Team, 2022) and the web scraping package rvest (Wickham, 2022) have been employed (further details are provided in section 3.6).

2.2.2 Maintenance Process

To carry out the sixth maintenance of the FoodEx2 catalogue, the TG performed the following steps:

- Evaluation of the requests, comments and proposals from internal and external stakeholders
- Decision on the implementation of the proposals based on the following options: addition of new terms, extension or better specification of the scope of existing terms or identification of the correct way of implementing the suggestion using already existing terms and rules, change of the parent-child relationship of the terms, deprecation and setting of terms as not reportable/dismissed in one hierarchy but still reportable in others.
- Checking for inconsistencies and ambiguities
- Update of the FoodEx2 catalogue
- Summary of the changes for the present technical report.

3 Maintenance Output

Figure 1 summarizes the quantities of different maintenance operations performed in the maintenance window of 2022.

In the following sections, the details on these maintenance operations and rationale behind these changes will be discussed. This overview of the introduced changes is mainly structured along the impacted food safety domains and stakeholders, and legislative changes necessitating these modifications. We will conclude with additional cross-domain overviews on specific maintenance operations.

A detailed table with all additions performed during 2022, the hierarchies and facets affected and the rationale behind is available in Appendix A.

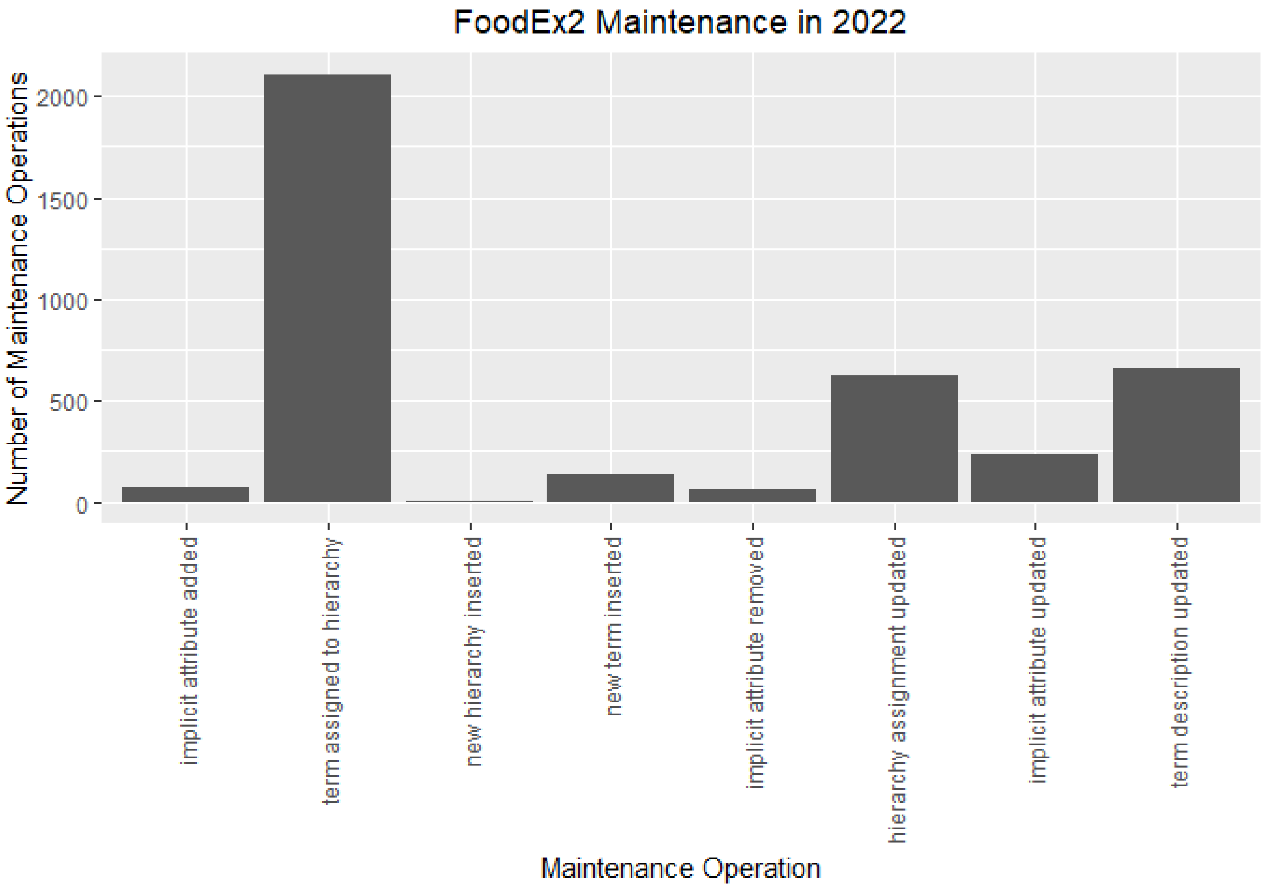

Figure 1: Summary of the different maintenance operations performed in the 2022 maintenance window

3.1 FoodEx2 updates in accordance with Avian Influenza surveillance programmes

FoodEx2 is used for reporting data under the Avian Influenza (AI) surveillance programmes in poultry and wild birds, carried out by Member States, according to the Commission Decision 2010/367/EU[3] and guidelines provided by Council Directive 2005/94/EC[4] (EFSA, 2018a). To meet the data reporting requirements in this domain, extensive amendments to the Bird section, 'A057X – Bird (as animal)', have been performed in 2019 and 2020 (EFSA, 2020, 2021b), followed by additional minor additions and revisions during the maintenance process in 2021. In the maintenance window of 2022, a consolidation phase was finished, with the objective to harmonize the taxonomic sub-tree of this section with scientific state of the art knowledge on the taxonomic classification of Aves (birds), which recently has been undergoing revisions due to adoption of new techniques such as genomic sequence analysis.

This revision of the taxonomic structuring was conducted with the support of an external expert in ornithology engaged under a negotiated procedure by the Biological Hazards and Animal Welfare (BIOHAW) Unit (former Animal and Plant Heath (ALPHA)). The terms from the 'Bird (as animal)' section in the F01 – Source facet thus have been reviewed to verify the scientific and common names of all species and associated sources, standardize the taxonomy against globally

[3] Official Journal of the European Union, Commission Decision of 25 June 2010 on the implementation by Member States of surveillance programmes for avian influenza in poultry and wild birds (notified under document C(2010) 4190) (Text with EEA relevance) (2010/367/EU)

[4] COUNCIL DIRECTIVE 2005/94/EC of 20 December 2005 on Community measures for the control of avian influenza and repealing Directive 92/40/EEC

recognized systems, fill gaps in the taxonomic hierarchy (e.g., performing splitting operations on previous terms to fill gaps in the hierarchy, such as complementing terms on genus or species level), and standardize additional provided information (esp. external references, scope notes etc.). In addition, the literature was reviewed to identify those species potentially occurring naturally in the wild (i.e., including free-living species and excluding species held in collections, zoos etc.) and distinguishing them from domestic birds. As the primary taxonomic backbone, the IOC World Bird List (Gill et al., 2022), an open access resource of the international community of ornithologists, has been selected due to its comprehensiveness and regular maintenance. This primary resource was complemented with the Clements Checklist of Birds providing, in taxonomic sequence, up-to-date information (e.g., scientific name, English name, description of the worldwide range of each species and subspecies) on a wide range of bird species recognized by the scientific and birding communities. For instance, the implicit attributes of the terms from the 'Bird (as animal)' section, such as the scientific and English names, were also checked against the current Clements list (Clements et al., 2022). Moreover, the Global Red List published by IUCN (HBW and BirdLife International, 2022) has been selected since it is regularly reviewed and updated according to a robust consultation process.

As a result of this activity, during the maintenance of 2022, a total number of 190 changes were implemented in terms of common name, scientific name/synonym, term name, scope notes, parent-child relationships and/or Euring codes.

Due to this taxonomy hierarchy revision and to accommodate data providers' requests for the Avian Influenza data collection, n = 24 new birds species have been added and made reportable in the F01 – Source facet in the MTX 14.0 release. The newly added taxonomic entities included two orders, three families, three genera, fifteen species and one subspecies. Most newly added terms were complemented with their relevant Euring code[5].

This systematic revision also motivated changes in existing terms, which were associated with: 1) update/addition of new common/scientific names, 2) update of the term extended name, 3) removal of existing common names and/or scientific names to ensure more accurate description of the term names, and 4) alterations to the order in which scientific names were originally presented. Term extended names were updated for only one code, common names were added or updated for eight terms, while scientific name/synonyms were added or updated for 73 terms. A new parent term was assigned to 21 codes due to the updated taxonomic structure. The Euring codes specified in the EFSA catalogue were checked to ensure that they matched the respective taxonomic entity, using the publicly available updated list[6], and employed for the validation of species names present in the catalogue. The Euring code for the term 'A172A - Soft-plumaged Petrel (as animal)' was changed, since the original species represented by the old Euring code does not exist, rather it is represented by four individual species. Consequently, scope notes were updated for 86 terms, mostly to reflect changes in common and/or scientific names.

Three terms were dismissed from the 'A057X - Bird (as animal)' section in the F01 – Source facet: 'A172X - Pallas's Cormorant (as animal)' was dismissed since being extinct since the 1880s. The two other dismissals were related to the removal of duplicate terms: 'A058Q - Snipe (as animal)' was dismissed from the catalogue since being a duplicate to 'A188G - Sandpipers (as animal)'. Consequently, the term 'A188G - Sandpipers (as animal)' was updated in terms of extended name and scope notes to accommodate the aforementioned amendment (representing the category 'Sandpipers and Snipes (as animal)'). Moreover, the term 'A058R - Grouse (as animal)' was dismissed from the 'A057X - Bird (as animal)' section in the F01 – Source facet, since being a duplicate to and thus replaced by the term 'A17VM - Willow Ptarmigan (as animal)'. Hence, the scientific name and scope note of the term 'A01TJ - Grouse fresh meat' were updated due to this dismissal of its previous implicit F01 -Source facet ('A058R- Grouse (as animal)') and

[5] Euring is the coordinating organisation for European bird ringing schemes which maintains a databank containing ringing recovery information gathered throughout Europe.
[6] https://app.bto.org/euringcodes/species.jsp

replaced by the unified term 'A17VM - Willow Ptarmigan (as animal)'. Thus, these modifications also led to the update of scope notes and/or scientific name for 2 terms that are reportable in Exposure and other hierarchies.

3.2 Revisions in the Feed hierarchy

The Feed hierarchy was revised according to the latest version of the Catalogue of feed materials[7]. Since the last revision of the Catalogue of feed materials pursuant to Commission Regulation (EU) No 68/2013 published 2017, the appropriate representatives of the European feed business sectors have, in consultation with other parties concerned, in collaboration with the competent national authorities and taking into account relevant experience from opinions issued by the European Food Safety Authority and scientific or technological developments, developed amendments to the Catalogue of feed materials. Those amendments refer to clarifications of the general provisions, new entries for treatment processes and for feed materials as well as adaptations of existing entries.

As a result, changes on 67 terms were implemented in the Feed hierarchy in order to meet the new legislative demands, including addition of new terms, dismission of existing terms and changes in term extended name, scope notes, parent-child relationship as well as in the area of implicit attributes or facets. The majority of these amendments was associated with the addition of new feed materials, followed by the dismission of terms and the update on existing terms in relation to the extended name, scope notes, parent-child relationship, implicit attributes or facets.

The "Feed section" ('A0BB9 – Feed') of the Reporting hierarchy was updated according to the Commission Regulation (EU) 2022/1104 amending Commission Regulation (EU) No 68/2013 on the Catalogue of feed materials[7]. As a result of this review, 33 new terms were added to the Feed hierarchy. New terms were added to almost all of the broader categories present in the Feed hierarchy. Most of them (n = 9) were added to the category 'A0BRA - Fermentation (by-)products from microorganisms the cells of which have been inactivated or killed (feed)', which contains feed materials such as fermentation products obtained from whole micro-organisms or their parts, fermentation co-products mainly consisting of microbial biomass and other fermentation co-products. New feed materials were also added to the following categories in the Feed hierarchy: 'A0BBA - Cereal grains and products derived thereof (feed)' (n=4), 'A0BEB - Oil seeds, oil fruits, and products derived thereof (feed)' (n=5), 'A0BG2 - Legume seeds and products derived thereof (feed)' (n=1), 'A0BH7- Tubers, roots, and products derived thereof (feed)' (n=2), 'A0BKT- Forages and roughage, and products derived thereof (feed)' (n=2), 'A0BLK - Other plants, algae, funghi and products derived thereof (feed)' (n=3), 'A0BMY - Land animal products and products derived thereof (feed)' (n=1), 'A0BNJ- Fish, other aquatic animals and products derived thereof (feed)' (n=1), 'A0BP4 -Minerals and products derived thereof (feed)' (n=4), 'A0BRR - Miscellaneous (feed)' (n=1). "EUFeedReg", an implicit attribute that was previously created to keep track of the assigned number of terms from the Catalogue of Feed Materials, was added to all new terms that originated from this source. An additional enhancement of the Feed section was the implementation of implicit facet F27 - Source-commodity. Apart from implicit facets F02 – Part-nature and F23 – Target-consumer that are already assigned to all feed materials, two new Raw Primary Commodity (RPC) terms have been added (made reportable) in the F27 – Source-commodities facet in order to be assigned as implicit Source-commodities to their derivatives. Similarly, seven newly added RPC derivatives have implicitly assigned an F27 –Source-commodity facet being the correspondent RPC term.

Taking into account the amendments in the Feed area, term extended name, scope notes, parent-child relationship and implicit facets or attributes (e.g., "EUFeedReg") were updated for 17 codes. Most of those amendments concern the update of the term extended name and scope notes to better specify the scope of the terms. For instance, apart from animal by-products, the

[7] Official Journal of the European Union, Commission Regulation (EU) 2022/1104 amending Regulation (EU) No 68/2013 on the Catalogue of feed materials, https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:32022R1104&from=EN

word 'product' or 'co-product' in scope notes, as appropriate, is used instead of the word 'by-product' to reflect the market situation and the language used in practice by feed business operators to highlight the commercial value of feed materials.

As shown in Table 3 – List of non-reportable terms during the maintenance in 2022, 17 terms have been dismissed from the Feed hierarchy, in accordance with the latest version of the Catalogue of Feed materials[7]. These 17 terms were dismissed from the Reporting hierarchy, too, to align the way of reporting between different domains. Most terms (n=16) were dismissed from the category 'A0BRA - Fermentation (by-)products from microorganisms the cells of which have been inactivated or killed (feed)'. Nevertheless, the information related to most dismissed terms can be incorporated to the newly added terms within this category. For instance, since the term 'A0BRD - Protein from Methylococcus capsulatus (Bath), Alca ligenes acidovorans, Bacillus brevis and Bacillus firmus (feed)' has been dismissed from the hierarchy, the newly added term 'A18XN - Single cell proteins from bacteria (feed)' can be used instead. In particular, according to the Commission Regulation (EU) 2022/1104, the term 'A18XN - Single cell proteins from bacteria (feed)' includes the protein products obtained by fermentation with bacteria on a substrate/culture medium consisting of methanol (fermented with Methylophilus methylotrophus) or natural gas (fermented with Methylococcus capsulatus, Alcaligenes acidovorans, Aneurinibacillus danicus (previously known as Bacillus brevis) and/or Bacillus firmus) as carbon source, a nitrogen source of vegetal or chemical origin, vitamins and minerals. Thus, in case of reporting the protein from Methylococcus capsulatus using the new term 'A18XN - Single cell proteins from bacteria (feed)', the species of microorganism(s) included in this category shall be indicated with the name of the feed material (i.e., "name as in the catalogue" + "name of the species"), such as "Single cell proteins from Methylococcus capsulatus". A similar approach should be followed for other species such as "Methylophilus methylotrophus", for which the new term 'A0BRC - Protein from Methylophilus methylotrophus (feed)' has been dismissed, or whenever the new term 'A18XP - Inactivated bacteria and parts thereof (feed)' should be used (e.g., "Inactivated Lactobacillus acidophilus"), accordingly. Moreover, one term was dismissed from the category 'A0BLK - Other plants, algae, funghi and products derived thereof (feed)' due to the new legislative demands.

3.3 Terms requested by FAO/WHO

41 terms have been added based on requests from the Food and Agricultural Organisation (FAO) / World Health Organization (WHO) for mapping the food consumption data within the Global Individual Food consumption data Tool (GIFT). Most of these changes affect the inclusion of novel plant ('as plant') and animal ('as animal') species, which were made reportable in the F01 – Source facet, as captured in

Table 2 - Update of the F01 source facet hierarchy during the maintenance in 2022.

If the requested terms were already present in the catalogue under a different name or scientific name, the addition of these terms to the catalogue was not taken into consideration. Instead, the existing term was amended with the common name or synonym of the scientific name and coding solutions to the users were provided. Furthermore, in some cases, the addition of requested terms was postponed to the next release, when further specifications were needed before a term was decided to be added.

Moreover, the mapping of FoodEx2 terms to their corresponding classification in WHO's GEMS (Global Environment Monitoring System / Food Contamination Monitoring and Assessment Programme, commonly known as GEMS/Food) - classification system was updated in-line with WHO's GEMS update performed in December 2021[8]. Thus, the implicit attribute "GEMSCode" was updated for 202 terms.

[8] https://cdn.who.int/media/docs/default-source/food-safety/gems-food/gems-instructions-for-electronic-submission-of-data.pdf?sfvrsn=c79dd32c_7

3.4 Accommodating the novel Food Contact Materials Data Collection

In accordance with Article 29(1)(a) of Regulation (EC) No 178/2002, EFSA has received a mandate from the European Commission to conduct preparatory work for the re-evaluation of phthalates, structurally similar substances and replacement substances that are potentially used as plasticisers in food contact materials (FCMs), to consider migration of plasticisers from FCMs in the context of the dietary exposure assessment. Thus, EFSA has launched a dedicated novel data collection on phthalates, structurally similar substances and replacement substances migrating from or occurring in food contact materials in 2022[9]. To accommodate the needs of this new data collection, Facet F19 – Packaging-material thus has been substantially expanded, leading to the inclusion of 33 new terms in the MTX/packmat hierarchies, to allow for a fine-grained recording of the packaging material used.

3.5 Changes in the Pesticides domain

Two types of changes were induced by the domain of pesticides data collection and analysis: The introduction of a novel hierarchy intended for the PRIMo (Pesticide Residue Intake Model) tool, and a range of updates on the implicit matrix code attribute.

3.5.1 Introduction of the PRIMo (Pesticide Residue Intake Model) hierarchy

Current version of the Pesticide Residue Intake Model (PRIMo version 3.1) (EFSA, 2018b) is the standard tool employed at EU level for the dietary risk assessment for pesticide residues in the framework of setting and reviewing of maximum residue levels for pesticides under Regulation (EC) No 396/2005 and in the peer review of pesticides under Regulation (EU) No 1107/2009. Release 13.2 introduced the novel PRIMo hierarchy, providing a grouping of the FoodEx2 terms according to Part A and Part B codes of Annex I of Regulation (EC) 396/2005, for use in the upcoming new version of EFSA's PRIMo tool (revision 4). As opposed to the current Excel-based spreadsheets, the new PRIMo4 will be an online tool introducing a new user interface and various improvements, such as the integration of new consumption data and introduction of the PRIMo hierarchy in the calculations. 1901 terms have been added to this new hierarchy.

3.5.2 Updates on the matrix code attribute

The catalogue was also revised in line with Annex I of the Regulation (EC) No 396/2005 related to pesticides[10]. 40 terms received an update of their implicit attribute "matrixCode", which associates a FoodEx2 term with its corresponding legislative category specifying its applicable Maximum Residue Limit (MRL). 17 matrix codes have been removed and 23 matrix codes have been updated in the 2022 FoodEx2 maintenance window.

3.6 Updates in the Plants section

Throughout the 2022 maintenance window, 660 term descriptions (i.e., scope notes) have been updated. The largest share of this updating of scope notes (n = 529) was performed in the Botanicals hierarchy, i.e., the branch of the 'Plants (as plant)' section in the F01 – Source facet. This was since one of the major external references used – The Plant List (TPL)[11], an online taxonomic database of all known plant species produced by the botanical community, created in a collaboration between the Royal Botanic Gardens, Kew and Missouri Botanical Garden – has been deprecated and superseded by World Flora Online (WFO)[12], an open-access, Web-based compendium of the world's plant species. WFO has been developed by a consortium of leading botanical institutions worldwide in response to the "2011-2020 GSPC's updated Target 1: to

[9] https://www.efsa.europa.eu/en/call/call-collection-data-phthalates-structurally-similar-substances-and-replacement-substances

[10] Official Journal of the European Union, Commission Regulation (EU) 2018/62, replacing Annex I to Regulation (EC) No 396/2005 of the European Parliament and of the Council https://eur-lex.europa.eu/legal-content/EN/TXT/HTML/?uri=CELEX:32018R0062&from=EN#ntr12-L_2018018EN.01000302-E0012

[11] http://theplantlist.org/

[12] http://www.worldfloraonline.org/

achieve an online Flora of all known plants by 2020", and its Taxonomic Backbone is actively curated by the community of WFO Taxonomic Expert Networks (TENs). Using R (R Core Team, 2022) and the web scraping package rvest (Wickham, 2022), 529 terms having an external reference to http://theplantlist.org/ have been identified from the Excel spreadsheet export of the catalogue, and the URL to their corresponding up-to-date WFO page has been extracted and updated, or manually revised in the case of HTTP errors due to outdated URLs (42 terms, mostly on those terms in which TPL has been the sole provided reference).

Other changes

3.6.1 Changes in the F28 – Process facet

Based on a request from FAO, the new term 'A18TD - Air-frying' has been added to the F28 – Process facet (see Figure 2) under 'A0BA1 - Cooking and similar thermal preparation processes', describing the cooking process of simulating deep frying without submerging the food in oil by circulating hot air at high speed, a cooking technique that has recently gained popularity.

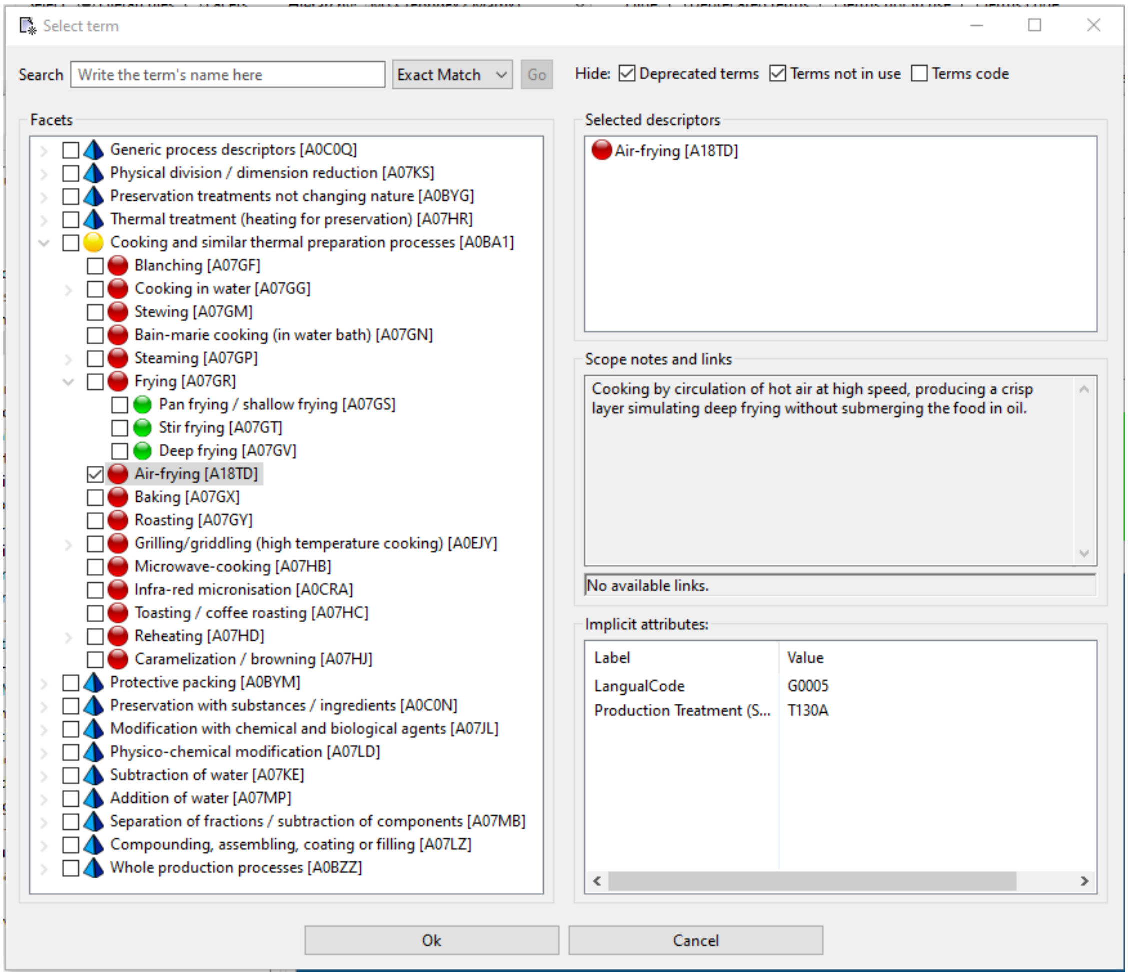

Figure 2: Current view of the F28 - Process facet – screenshot from the FoodEx2 catalogue browser

3.6.2 Changes in the F21 – Production-method facet

To accommodate the needs for zoonoses monitoring activities, specifically for the reporting of *Trichinella*[13] testing of pigs, Facet F21 – Production method has been refined. The term 'A07SF - Raised under controlled housing conditions (Reg. (EU) 2015/1375)' has been dismissed and

[13] EFSA (European Food Safety Authority), 2022. Scientific Network for Zoonoses Monitoring Data Minutes of the 40th meeting. Available online: https://www.efsa.europa.eu/sites/default/files/2022-10/221013_14-m.pdf

replaced by the following two, more specific terms that should be chosen by data providers instead to indicate whether recognition of controlled housing conditions was performed by the competent authority or not (i.e., farming conditions where animals are kept at all times under conditions controlled by the food business operator with regard to feeding and housing) (see Figure 3):

- 'A18YL - Raised under controlled housing conditions, recognised by the competent authorities'
- 'A18YM - Raised under controlled housing conditions, not recognised by the competent authorities'

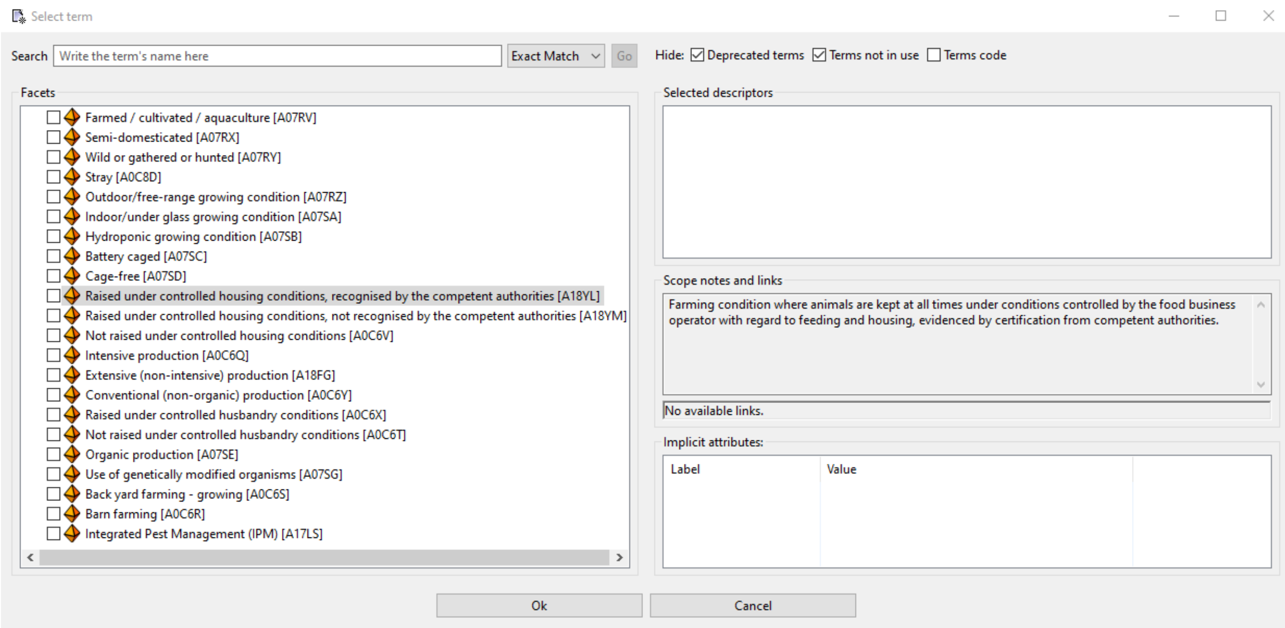

Figure 3: Current view of the F21 - Production-method facet – screenshot from the FoodEx2 catalogue browser

3.6.3 Changes in the Animal fresh meat - Amphibians and Reptiles section

In view of new regulations with the regard to requirements for the entry into the Union of consignments of food-producing animals and certain goods intended for human consumption[14] and on the performance of official controls on the use of pharmacologically active substances authorised as veterinary medicinal products or as feed additives and of prohibited or unauthorised pharmacologically active substances and residues thereof[15,16], a separate product category for reptiles was requested for the monitoring of Veterinary Medicinal Product Residue (VMPR) substances. Thus, the children of term 'A02KQ - Amphibians and Reptiles' have been restructured in the major (MTX) as well as derived hierarchies (i.e., Reporting hierarchy, Zoonoses hierarchy, Exposure hierarchy, Veterinary Drug Residues hierarchy, Source-

[14] Official Journal of the European Union, Commission Delegated Regulation (EU) 2022/2292 supplementing Regulation (EU) 2017/625 of the European Parliament and of the Council with regard to requirements for the entry into the Union of consignments of food-producing animals and certain goods intended for human consumption, https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:32022R2292

[15] Official Journal of the European Union, Commission Delegated Regulation (EU) 2022/1644 supplementing Regulation (EU) 2017/625 of the European Parliament and of the Council with specific requirements for the performance of official controls on the use of pharmacologically active substances authorised as veterinary medicinal products or as feed additives and of prohibited or unauthorised pharmacologically active substances and residues thereof, https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:32022R1644&from=EN

[16] Official Journal of the European Union, Commission Implemented Regulation (EU) 2022/1646 on uniform practical arrangements for the performance of official controls as regards the use of pharmacologically active substances authorised as veterinary medicinal products or as feed additives and of prohibited or unauthorised pharmacologically active substances and residues thereof, on specific content of multi-annual national control plans and specific arrangements for their preparation, https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:32022R1646&qid=1664547039710&from=EN

commodities and Ingredient hierarchy), as indicated in Figure 4, by introducing two dedicated higher-level parent terms – 'A18YJ - Amphibians' and 'A18YK - Reptiles'. To keep this branch in alignment with the categorisation defined in Regulation (EC) No 396/2005 ("EU pesticide regulation")[10], term 'A0CVD - Other Amphibians and Reptiles' needed to be retained as separate term to match the category "Other Amphibians and Reptiles" with associated matrix code P1050000-990 as specified in Annex I, Part A ("Products of plant and animal origin referred to in Article 2(1) to which MRLs apply") of Regulation (EC) No 396/2005.

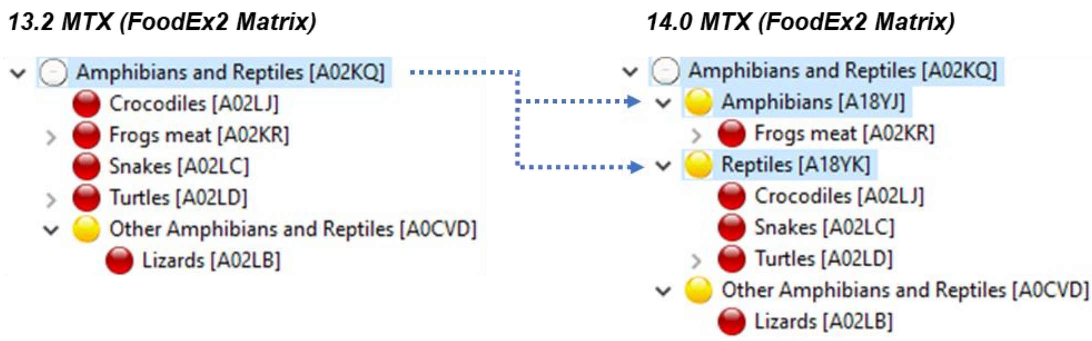

Figure 4: Revision of the "Amphibians and Reptiles" section - changes compared to the previous catalogue version

3.7 Amendments to implicit facets

Table 1 - Updates on implicit facets during the maintenance in 2022

FoodEx2Code	Term Extended Name	Old implicit facet	New implicit facet
A01TH	Snipe fresh meat	F01.A058Q$F27.A01TH	F01.A188G$F27.A01TH
A01TJ	Grouse fresh meat	F01.A058R$F27.A01TJ	F01.A17VM$F27.A01TJ
A031K	Quail eggs	F01.A058J$F27.A031K	F01.A18MV$F27.A031K
A03TF	Textured soy protein	F27.A015P	F04.A015P
A03TG	Textured fungal proteins	F27.A049B	F04.A049B
A0BHK	Fructo-oligosaccharides (feed)	F27.A05JE	

Implicit facets were updated for six terms available in different hierarchies and facets after being recognized as insufficient or incorrect by TG or data providers. In the Feed hierarchy, implicit facets were updated for only one term. In particular, for the term 'A0BHK - Fructo-oligosaccharides (feed)' the F27 – Source-commodities facet was removed since this material can be obtained from sugar beet or sugar cane, and not only from sugar beet as indicated by the old one. In addition, as presented in Table 1, for the terms 'A01TH -Snipe fresh meat' and 'A01TJ - Grouse fresh meat' implicit facet F01 – Source was changed since the old ones 'A058Q - Snipe (as animal)' and 'A058R- Grouse (as animal)', respectively, were dismissed from the 'A057X - Birds (as animal)' section. Moreover, implicit facet F01 –Source has changed for the term 'A031K - Quail eggs' in order to describe better the parent-child relationships as it represents the eggs from the animal classified under the genus Coturnix.

3.8 Amendment of the F01 – Source facet in accordance with the ongoing projects and data collections

Table 2 - Update of the F01 source facet hierarchy during the maintenance in 2022

FoodEx2 Code	Term Extended Name	Source Parent Code Old	Source Parent Code
A058F	Partridge (as animal)	A0CNF	A188Z
A058G	Pheasant (as animal)	A188Z	A18TT
A058J	Quail (as animal)	A0CNF	A18MV
A0CNG	Bobwhite (as animal)	A058F	A18YH
A0CNH	Common Quail (as animal)	A058J	A18MV
A0CNJ	Japanese Quail (as animal)	A058J	A18MV
A13KH	Dwarf copperleaf (as plant)		A05DY
A140P	Celosia (as plant)		A05DY
A14HR	Erythroxylum coca Lam. (as plant)		A065G
A157Y	African basil (as plant)		A05MX
A159V	Common screw pine (as plant)		A061Q
A15MJ	Katuk (as plant)		A05DY
A15RK	Jurubeba (as plant)		A061Q
A172C	Scopoli's Shearwater (as animal)	A187V	A18YE
A174K	Great Bittern (as animal)	A18JM	A18ME
A176K	Alligator weed (as plant)		A05DY
A17QB	Grey Partridge (as animal)	A188Z	A189A
A17RM	Hooded crow (as animal)	A17RN	A17QD
A17YH	Mediterranean Gull (as animal)	A0CST	A18LF
A180X	Water Pipit (nominate) (as animal)	A181A	A180V
A184G	Carrion Crow (nominate) (as animal)	A17RN	A17QD
A185P	European robin (nominate) (as animal)	A18PQ	A17NM
A188Z	Pheasants and Allies (as animal)	A058J	A0CNF
A189R	Gran Canaria Robin (marionae) (as animal)	A18PQ	A17NM
A189V	Audouin's Gull (as animal)	A058T	A18LF
A189Y	Pallas's Gull (as animal)	A058T	A18LF
A18AB	White Wagtail (nominate) (as animal)	A181E	A181D
A18FM	Rock Pigeon (nominate) (as animal)	A17RL	A058H
A18MH	Green pheasant (as animal)	A188Z	A18TT

F01 – Source facet was amended with 89 new terms to allow mapping of samples within different data collections and projects. In particular, given the extensive amendment of the Bird section, 'A057X –Bird (as animal)', during the maintenance process in 2022, an additional structuring in terms of completion of taxonomy and parent-child relationship was required. Consequently, the addition of new terms requested by data providers led to the update of the parent for existing terms in the catalogue. For instance, the addition of 'A18YF - Cory's Shearwater (as animal)' led to the creation of a new genus. As a result, a new parent term was assigned to the newly added term 'A18YF - Cory's Shearwater (as animal)' and the already existing term 'A172C - Scopoli's Shearwater (as animal)', as well.

In addition, the update of the taxonomic structure in terms of completing the parent-child relationships was implemented on many existing terms which represent a subspecies level. Consequently, in these cases, the relevant species present in the catalogue was assigned as new parent terms instead of the higher-level taxonomic entity (e.g., 'Carrion Crow (nominate) (as animal)', 'White Wagtail (nominate) (as animal)' etc.). The update of the parent-child relationships was also implemented due to the review conducted in order to identify those species potentially occurring naturally in the wild (i.e., including free-living species and excluding species held in collections, zoos etc.) and distinguishing them from domestic birds. In particular, among other changes, the taxonomic structure was updated for the terms 'A058F - Partridge (as animal)' and 'A058J -Quail (as animal)', since these groupings were identified to represent domestic birds. For instance, scope notes, implicit attributes and parent-child relationship for the term 'A058J - Quail (as animal)' were updated to accommodate the fact that the particular grouping is specific to the domestic quail and originally descending from some taxonomic Coturnix species (e.g., Coturnix japonica or Coturnix coturnix). A similar approach was also implemented for the term 'A058F - Partridge (as animal)'.

3.9 Deprecated and dismissed terms during the maintenance 2022

During the maintenance window of 2022, no term has been deprecated, but several terms have been dismissed from the Reporting hierarchy, thus, cannot be used anymore for describing samples in up-coming data collections. Table 3 – List of non-reportable terms during the maintenance in 2022 summarizes all implemented changes in reportability. As described in the corresponding domain-specific sections, most dismissed terms have been replaced by a novel term that should be used instead. Please see the corresponding domain-specific sections for further details.

Conclusion

The present report documents the update of FoodEx2 performed within three minor releases published in 2022 (MTX 13.1 and MTX 13.2) and the major release published in January 2023 (MTX 14.0), covering requests from internal and external users.

- In total, 137 new terms have been added to the catalogue since the last major release 13.0.
- Throughout the 2022 maintenance window, 660 scope notes have been updated, to keep their term descriptions and references up-to-date with current scientific knowledge and resources as well as legislative definitions.
- One new hierarchy has been included (PRIMo).
- 2108 terms have received a new hierarchy assignment, 620 hierarchy assignments have been updated.
- 65 new implicit attributes have been added, 64 implicit attributes have been removed and 238 implicit attributes have been updated.

The complete content of the FoodEx2 system is available in Appendix B to this document as exported spreadsheet MTX_14.0.xlsx.

Table 3 – List of non-reportable terms during the maintenance in 2022

FoodEx2 Code	FoodEx2 Term name	Hierarchy in which the term was dismissed
A01JF	Chinese olives, black, white	biomo - exposure - ingredient- racsource - report -
A058Q	Snipe (as animal)	source
A058R	Grouse (as animal)	source
A172X	Pallas's Cormorant (as animal)	source
A07SF	Raised under controlled housing conditions (Reg. (EU) 2015/1375)	master
A0BLQ	Algae extract; [Algae fraction] (feed)	feed - report
A0BRB	Products obtained from the biomass of specific micro-organisms grown on certain substrates (feed)	feed - report
A0BRC	Protein from Methylophilus methylotrophus (feed)	feed - report
A0BRD	Protein from Methylococcus capsulatus (Bath), Alca ligenes acidovorans, Bacillus brevis and Bacillus firmus (feed)	feed - report
A0BRE	Bacterial protein from Escherichia coli (feed)	feed - report
A0BRF	Bacterial protein from Corynebacterium glutamicum (feed)	feed - report
A0BRH	Yeasts from biodiesel process (feed)	feed - report
A0BRJ	Other fermentation by-products (feed)	feed - report
A0BRL	By-products from the production of L-glutamic acid (feed)	feed - report
A0BRM	By-products from the production of L-lysine-monohydrochloride with Brevibacterium lactofermentum (feed)	feed - report
A0BRN	By-products from the production of amino acids with Corynbacterium glutamicum (feed)	feed - report
A0BRP	By-products from the production of amino acids with Escherichia coli K12 (feed)	feed - report
A0BRQ	By-product of enzyme production with Aspergillus niger (feed)	feed - report
A0EGP	Mycelium silage from the production of penicillin (feed)	feed - report
A186T	Product from Lactobacillus species rich in protein (feed)	feed - report
A186Y	Product from Aspergillus oryzae rich in protein (feed)	feed - report
A186Z	Polyhydroxybutyrate from fermentation with Ralstonia eutropha (feed)	feed - report

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Abbreviations

AI	Avian Influenza
ALPHA	Animal and Plant Health
BIOHAW	Biological Hazards and Animal Welfare
EC	European Commission
EFSA	European Food Safety Authority
EU	European Union
FAO	Food and Agriculture Organization
FCM	Food Contact Material
FoodEx2	Food Classification and Description System
GEMS	Global Environment Monitoring System
GIFT	Global Individual Food consumption data Tool
HTTP	Hypertext Transfer Protocol
IDATA	Integrated Data
MRL	Maximum Residue Limit
MTX	FoodEx2 Matrix hierarchy
PRIMo	Pesticides Residue Intake Model
RPC	Raw Primary Commodity
TG	Technical Group
TPL	The Plant List
URL	Uniform Resource Locator
VMPR	Veterinary Medicinal Product Residue
WFO	World Flora Online
WHO	World Health Organization

Appendix A – New terms added during the 2022 maintenance

Appendix B – FoodEx2 catalogue MTX_14.0